

on the annealing behavior of voids.

IX.C.4.a. Analysis and Results

When irradiation is turned off, as in post irradiation annealing experiments, or in the annealing mode of voids in between pulses, thermal vacancies are emitted from different components of the microstructure and absorbed by others. If the time between pulses is longer than a vacancy mean lifetime, or in a post irradiation annealing experiment, a simple point defect balance situation can be described. Since thermal emission of interstitials from microstructures has a low probability (because of the large interstitial formation energy), a balance equation for vacancies, a long time after the irradiation is turned off, can be described as:

$$\left[\begin{array}{l} \text{Total rate of thermal vacancy emission} = \text{total rate} \\ \text{of vacancy removal} \end{array} \right]$$

Since thermal concentration of interstitials is very low under these circumstances, vacancy removal by mutual recombination is negligible, and the previous balance equation can be written as:

$$P^e = P_{sv} = \lambda_v \bar{c}_v . \quad (9.18)$$

Therefore, $\bar{c}_v = \frac{P^e}{\lambda_v} , \quad \text{or} \quad (9.19)$

$$\bar{c}_v = c_v^e \left\{ \frac{4\pi R_c N_c \exp\left\{\left(\frac{2\gamma}{R_c} - p_g\right)\frac{\Omega}{kT}\right\} + Z_{vd}^{a,ea} \exp(\sigma\Omega/kT) + Z_{vd}^{n,en} +}{4\pi R_c N_c + Z_{vd}^{a,ea} + Z_{vd}^{n,en} +} \right. \\
\left. \frac{Z_{vd}^{a,ila} \exp(\sigma\Omega/kT) \exp\left(-\frac{\{\gamma_{sf} + F_{el}(r_{il}^a)\}b^2}{kT}\right) + Z_{vd}^{n,iln} \exp\left(-\frac{\{\gamma_{sf} + F_{el}(r_{il}^n)\}b^2}{kT}\right) +}{Z_{vd}^{a,ila} + Z_{vd}^{n,iln} +} \right. \\
\left. \frac{Z_{vd}^{a,vla} \exp(\sigma\Omega/kT) \exp\left(-\frac{\{\gamma_{sf} + F_{el}(r_{vl}^a)\}b^2}{kT}\right) + Z_{vd}^{n,vln} \exp\left(-\frac{\{\gamma_{sf} + F_{el}(r_{vl}^n)\}b^2}{kT}\right) +}{Z_{vd}^{a,vla} + Z_{vd}^{n,vln} +} \right\} \quad (9.20)$$

Equation (9.20) shows that the average thermal vacancy concentration is a function of the microstructure and stress. The annealing of voids depends on the vacancy concentration gradient between the surface of the void and the average thermal vacancy concentration in the matrix. This gradient is proportional to the difference in concentrations as shown in Figure (9.16). According to this schematic representation we have one of 5 different situations:

Case	Average Thermal Vacancy Concentration Near Defects
1. Edge dislocations or specimen surfaces only	C_v^e
2. Small interstitial loops only	$C_v^e \exp\left\{-(F_{el} + \gamma_{sf})\frac{b^2}{kT}\right\}$
3. Small voids only	$C_v^e \exp\left(\frac{2\gamma}{R_c} - p\right)$
4. Small vacancy loops only	$C_v^e \exp\left\{+(F_{el} + \gamma_{sf})\frac{b^2}{kT}\right\}$
5. A mixed microstructure	$\bar{C}_v$

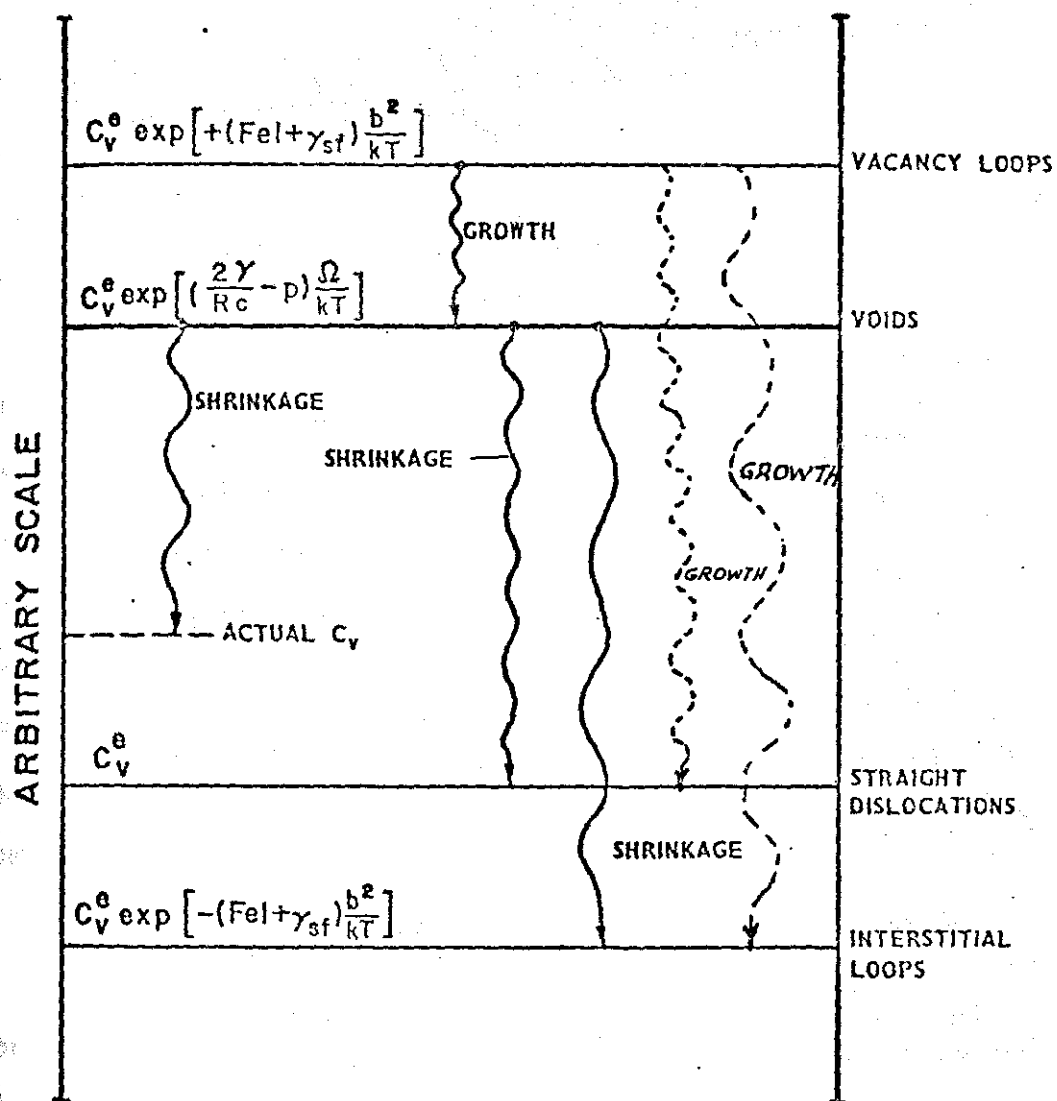


Fig. (9.16). Schematic illustration of main thermal annealing kinetics and their relation to microstructure.

One can see from Figure (9.16) that vacancy loops emit vacancies helping voids to grow as well as causing interstitial loops and dislocations to shrink) while smaller interstitial loops absorb more vacancies than edge dislocations, and therefore, make voids shrink faster.

To study the effect of changing the microstructure on the annealing of voids, the number of voids was fixed to 10^6 voids/cm³ and their radius to 152 Å. An equivalent number of interstitials to that of vacancies in the voids was distributed on interstitial loops. The only variable in the calculations was the ratio of interstitial loops to voids ranging from 10^{-6} to 10^4 with a negligible concentration of straight edge dislocations. It is evident from Figure (9.17) that voids anneal out much faster for larger ratios while the rate of annealing is slower when voids are the dominant thermal vacancy source/sink.

IX.C.4.b. Conclusions

Annealing of voids was shown to be a sensitive function of many variables, some variables enhance the annealing of voids, while others retard it. To enhance annealing of voids, one or more of the following conditions should exist:

- (1) High temperature.
- (2) Small void radius.
- (3) Large void surface energy.
- (4) High density of deformation produced dislocations.